SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4   **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I I. MD IMTHIAZ(192521360) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

I. IMTHIAZ

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]
- ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]
- ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]
- Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- P1: SoilDry → IrrigationNeeded
- P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
- P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
- Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- Stench [1,2] → Wumpus is adjacent to cell [1,2]
- Breeze [1,1] → Pit is adjacent to cell [1,1]
- Glitter [x,y] → Gold is at [x,y]
- ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

CASE STUDY 1 – Smart Medical Diagnosis System

(a) Representation using Propositional Logic

Let F = Patient has Fever, C = Patient has Cough, B = Patient has Breathlessness, R = Patient has Rash, Flu = Patient has Flu, M = Patient has Measles, and P = Patient has Pneumonia.

Rule 1: If the patient has Fever AND Cough, then the patient may have Flu.

```
F ∧ C → Flu
```

Rule 2: If the patient has Fever AND Rash, then the patient may have Measles.

```
F ∧ R → M
```

Rule 3: If the patient has Cough AND Breathlessness, then the patient may have Pneumonia.

```
C ∧ B → P
```

Facts about Patient A:

```
F = True
C = True
B = True
```

Therefore:

```
KB = {F, C, B, F ∧ C → Flu, F ∧ R → M, C ∧ B → P}
```

(b) Forward Chaining

Initial facts:

```
F = True
C = True
B = True
```

Step 1 – Apply Rule 1:

```
F ∧ C → Flu
F = True, C = True
Therefore: Flu = True
```

Derived fact: Patient A may have Flu.

Step 2 – Apply Rule 2:

```
F ∧ R → M
F = True, but R is not given.
Therefore Rule 2 cannot fire.
M = Not Derivable
```

Step 3 – Apply Rule 3:

```
C ∧ B → P
C = True, B = True
Therefore: P = True
```

Derived fact: Patient A may have Pneumonia.

Final conclusions:

```
Flu = True
Pneumonia = True
Measles = Not Derivable
```

(c) Backward Chaining for Pneumonia

Goal:

```
Pneumonia
```

Rule producing the goal:

```
Cough ∧ Breathlessness → Pneumonia
```

Goal reduction:

```
Pneumonia
   ↓
Cough ∧ Breathlessness
```

Subgoal 1:

```
Cough = True  ✓
```

Subgoal 2:

```
Breathlessness = True  ✓
```

Therefore:

```
Cough ∧ Breathlessness ⇒ Pneumonia
```

Goal Reduction Tree:

```
              Pneumonia
                  |
        Rule 3: Cough ∧ Breathlessness
             /              \
         Cough          Breathlessness
           |                  |
          TRUE               TRUE
```

Conclusion: Patient A has Pneumonia.

CASE STUDY 2 – Autonomous Traffic Management Agent

(a) Unification of Rule 1

Rule 1:

```
∀x (Vehicle(x) ∧ EmergencyType(x) → ClearPath(x))
```

Given facts:

```
Vehicle(Ambulance)
EmergencyType(Ambulance)
```

Matching Vehicle(x) with Vehicle(Ambulance):

```
x / Ambulance
```

Matching EmergencyType(x) with EmergencyType(Ambulance):

```
x / Ambulance
```

Most General Unifier:

```
θ = {x / Ambulance}
```

After substitution:

```
Vehicle(Ambulance) ∧ EmergencyType(Ambulance)
Therefore: ClearPath(Ambulance)
```

(b) Resolution Proof for Proceed(CarA)

Rule 1 in CNF:

```
¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)
```

Rule 2 in CNF:

```
¬ClearPath(x) ∨ GreenSignal(x)
¬ClearPath(x) ∨ ¬RedSignal(x)
```

Rule 3 in CNF:

```
¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)
```

Facts:

```
Vehicle(Ambulance)
EmergencyType(Ambulance)
Vehicle(CarA)
Behind(CarA, Ambulance)
```

Derive ClearPath(Ambulance):

```
¬Vehicle(Ambulance) ∨ ¬EmergencyType(Ambulance) ∨ ClearPath(Ambulance)
Vehicle(Ambulance)
```

```
¬EmergencyType(Ambulance) ∨ ClearPath(Ambulance)

EmergencyType(Ambulance)
```

```
ClearPath(Ambulance)
```

Derive GreenSignal(Ambulance):

```
¬ClearPath(Ambulance) ∨ GreenSignal(Ambulance)
ClearPath(Ambulance)
```

```
GreenSignal(Ambulance)
```

Negate the goal:

```
¬Proceed(CarA)
```

Instantiate Rule 3 with x = CarA and y = Ambulance:

```
¬Vehicle(CarA) ∨ ¬Behind(CarA,Ambulance) ∨ ¬GreenSignal(Ambulance) ∨ Proceed(CarA)
```

Resolve with Vehicle(CarA):

```
¬Behind(CarA,Ambulance) ∨ ¬GreenSignal(Ambulance) ∨ Proceed(CarA)
```

Resolve with Behind(CarA,Ambulance):

```
¬GreenSignal(Ambulance) ∨ Proceed(CarA)
```

Resolve with GreenSignal(Ambulance):

```
Proceed(CarA)
```

Resolve with the negated goal:

```
Proceed(CarA)
¬Proceed(CarA)
```

□

Conclusion: Proceed(CarA) is TRUE.

(c) Two Simultaneous Emergency Vehicles

The current knowledge base is not completely sufficient to handle two simultaneous emergency vehicles in all traffic situations.

Suppose:

```
Vehicle(Ambulance1)
EmergencyType(Ambulance1)
Vehicle(Ambulance2)
EmergencyType(Ambulance2)
```

Rule 1 can derive:

```
ClearPath(Ambulance1)
ClearPath(Ambulance2)
```

Rule 2 can derive:

```
GreenSignal(Ambulance1)
GreenSignal(Ambulance2)
```

However, the knowledge base does not contain rules for priority between two emergency vehicles, collision avoidance, or which direction receives the green signal.

Additional rules/facts needed include priority level, location and direction, route conflict information, priority selection, and collision avoidance.

```
EmergencyType(x) ∧ EmergencyType(y) ∧ x ≠ y → PriorityRequired(x,y)
HigherPriority(x,y) → GreenSignal(x)
GreenSignal(x) → ¬GreenSignal(y)  (when both compete for the same route)
```

Conclusion: The current KB handles individual emergency vehicles but not simultaneous conflicting emergencies completely.

CASE STUDY 3 – Agricultural Expert Reasoning System

(a) CNF Conversion

P1:

```
SoilDry → IrrigationNeeded
¬SoilDry ∨ IrrigationNeeded
```

CNF:

```
¬SoilDry ∨ IrrigationNeeded
```

P2:

```
IrrigationNeeded ∧ CropWheat → ApplyDripMethod
¬(IrrigationNeeded ∧ CropWheat) ∨ ApplyDripMethod
¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
```

CNF:

```
¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
```

P3:

```
¬ApplyDripMethod ∧ CropWheat → CropAtRisk
¬(¬ApplyDripMethod ∧ CropWheat) ∨ CropAtRisk
ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk
```

CNF:

```
ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk
```

Facts:

```
SoilDry
CropWheat
```

Final CNF:

```
{¬SoilDry ∨ IrrigationNeeded,
 ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod,
 ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk,
 SoilDry,
 CropWheat}
```

(b) Resolution Refutation for ApplyDripMethod

Goal:

```
ApplyDripMethod
```

Negate the goal:

```
¬ApplyDripMethod
```

Step 1:

```
¬SoilDry ∨ IrrigationNeeded
SoilDry
```

```
IrrigationNeeded
```

Step 2:

```
¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
IrrigationNeeded
```

```
¬CropWheat ∨ ApplyDripMethod
```

Step 3:

```
¬CropWheat ∨ ApplyDripMethod
CropWheat
```

```
ApplyDripMethod
```

Step 4 – Contradiction:

```
ApplyDripMethod
¬ApplyDripMethod
```

□

Resolution Proof Tree:

```
                         □
                       /   \
        ApplyDripMethod   ¬ApplyDripMethod
                 |
       ¬CropWheat
                 |
        CropWheat
                 |
     ¬IrrigationNeeded
                 |
      IrrigationNeeded
                 |
         SoilDry
```

Conclusion: ApplyDripMethod is logically proven.

(c) Remove SoilDry

Originally:

```
SoilDry → IrrigationNeeded → ApplyDripMethod
```

After removing SoilDry:

1. Rule P1 cannot fire because SoilDry is no longer available.
2. Therefore IrrigationNeeded cannot be derived.
3. Without IrrigationNeeded, Rule P2 cannot derive ApplyDripMethod.
4. P3 requires ¬ApplyDripMethod ∧ CropWheat. We know CropWheat, but we do not know ¬ApplyDripMethod.
5. Therefore CropAtRisk cannot be derived either.

Conclusion:

```
Without SoilDry, ApplyDripMethod cannot be concluded.
CropAtRisk also does not follow from the remaining KB.
```

This demonstrates the difference between not proving P and proving ¬P.

CASE STUDY 4 – Wumpus World Knowledge Agent

(a) Belief State and Modus Ponens

Given percepts:

```
Stench[1,2]
Breeze[1,1]
Glitter[2,2]
```

Rule 1:

```
Stench[1,2] → WumpusAdjacent[1,2]
```

By Modus Ponens:

```
WumpusAdjacent[1,2]
```

Rule 2:

```
Breeze[1,1] → PitAdjacent[1,1]
```

By Modus Ponens:

```
PitAdjacent[1,1]
```

Rule 3:

```
Glitter[x,y] → Gold[x,y]
```

Substitute x = 2, y = 2:

```
Gold[2,2]
```

Rule 4:

```
¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]
```

The rule requires explicit negative facts. The given percepts do not provide ¬Wumpus or ¬Pit for a specific cell, so this rule cannot establish safety.

Belief state:

```
{Stench[1,2],
 Breeze[1,1],
 Glitter[2,2],
 WumpusAdjacent[1,2],
 PitAdjacent[1,1],
 Gold[2,2]}
```

(b) Forward Chaining – Possible Wumpus and Pit Locations

Wumpus:

```
Stench[1,2]
       ↓
WumpusAdjacent[1,2]
```

Assuming standard four-directional adjacency, cells adjacent to [1,2] are:

```
[1,1], [1,3], [2,2]
```

Possible Wumpus locations:

```
[1,1], [1,3], [2,2]
```

Pit:

```
Breeze[1,1]
       ↓
PitAdjacent[1,1]
```

Adjacent cells:

```
[1,2], [2,1]
```

Possible Pit locations:

```
[1,2], [2,1]
```

Gold:

```
Glitter[2,2]
       ↓
Gold[2,2]
```

Therefore Gold is at [2,2].

(c) Safety of Path [1,1] → [1,2] → [2,2]

Proposed path:

```
[1,1] → [1,2] → [2,2]
```

Cell [1,1]:

```
Breeze[1,1]
       ↓
PitAdjacent[1,1]
```

```
       ↓
Possible pit locations: [1,2], [2,1]
```

Thus [1,2] cannot be proven safe from the given information.

Cell [1,2]:

```
Stench[1,2]
       ↓
WumpusAdjacent[1,2]
       ↓
Possible Wumpus locations: [1,1], [1,3], [2,2]
```

Thus the agent cannot prove that the relevant surrounding cells are safe.

Cell [2,2]:

```
Glitter[2,2]
       ↓
Gold[2,2]
```

The KB does not provide explicit ¬Wumpus[2,2] and ¬Pit[2,2]. Therefore Rule 4 cannot establish Safe[2,2].

Final decision:

```
The path [1,1] → [1,2] → [2,2] is NOT PROVABLY SAFE.
```

The agent should gather additional percepts or knowledge before proceeding.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand